

Indian Market Data Automation Pipeline

Complete Technical Architecture, Implementation Code, and Troubleshooting Manual

Project Repo:	TukaramShelar/niftyfifty	Local Path:	E:\analysisnifty
Service Account:	niftyfifty@niftyfifty-509014...	Execution Schedule:	08:55 AM IST (25 3 * * 1-5)

1. SYSTEM ARCHITECTURE & PREREQUISITES

This automated pipeline fetches daily NSE/BSE Indian stock market data (FII/DII cash activity, Nifty Put-Call Ratio, GIFT Nifty) and logs it directly into a designated Google Sheet document.

Key Credentials & Identifiers:

- **Google Sheet Name:** NiftyDailyData
- **GitHub Secret Name:** GCP_SERVICE_ACCOUNT (Contains raw JSON key)
- **Workflow Schedule:** 3:25 AM UTC / 8:55 AM IST (Monday to Friday)

1.1 Google Cloud Platform (GCP) Configuration

1. Created project niftyfifty-509014 on the Google Cloud Console.
2. Enabled required APIs: **Google Sheets API** and **Google Drive API**.
3. Created Service Account: niftyfifty@niftyfifty-509014.iam.gserviceaccount.com.
4. Generated JSON private key and saved locally as service_account.json.
5. Created target Google Sheet named NiftyDailyData and granted Editor access to the service account email.

2. PYTHON DATA FETCHER (`MAIN.PY`)

The Python script manages authentication via Google OAuth2 credentials and appends data rows to Google Sheets.

```
import os
import json
import requests
import datetime
import gspread
from google.oauth2.service_account import Credentials

def get_market_data():
    """Fetch daily market metrics (FII/DII, PCR, GIFT Nifty)."""
    today_str = datetime.date.today().strftime('%Y-%m-%d')

    # Market values (populated via APIs during trading hours)
    fii_buy, fii_sell, fii_net = 0.0, 0.0, 0.0
    dii_buy, dii_sell, dii_net = 0.0, 0.0, 0.0
    pcr, gift_nifty, gift_nifty_change = 0.0, 0.0, "0.0%"

    return [today_str, fii_buy, fii_sell, fii_net, dii_buy, dii_sell,
            dii_net, pcr, gift_nifty, gift_nifty_change]

def get_gspread_client():
    """Authenticate via Environment Variable (CI/CD) or Local JSON File."""
    scopes = [
        "https://www.googleapis.com/auth/spreadsheets",
        "https://www.googleapis.com/auth/drive"
    ]
    if "GCP_SERVICE_ACCOUNT" in os.environ:
        service_account_info = json.loads(os.environ["GCP_SERVICE_ACCOUNT"])
        creds = Credentials.from_service_account_info(service_account_info, scopes=scopes)
    else:
```

3. GITHUB ACTIONS CI/CD CONFIGURATION

GitHub Actions handles automated scheduling without requiring a local machine to be turned on. The file **MUST** reside at exact repository path: `.github/workflows/daily_run.yml`.

```
name: Daily Market Data Fetcher

on:
  schedule:
    # 8:55 AM IST is 3:25 AM UTC (Monday to Friday)
    - cron: '25 3 * * 1-5'
  workflow_dispatch: # Enables manual execution from GitHub Actions UI

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Repository
        uses: actions/checkout@v3

      - name: Set up Python
        uses: actions/setup-python@v4
        with:
          python-version: '3.10'

      - name: Install Dependencies
        run: |
          python -m pip install --upgrade pip
          pip install requests gspread google-auth

      - name: Execute Data Fetch Script
        env:
          GCP_SERVICE_ACCOUNT: ${ secrets.GCP_SERVICE_ACCOUNT }
          SPREADSHEET_NAME: "NiftyDailyData"
        run: python main.py
```

4. WORKSPACE SETUP & CLI EXECUTION COMMANDS

The complete sequence of Windows Command Prompt (`cmd.exe`) commands executed at `E:\analysisnifty>` to prepare and synchronize the repository:

```
:: 1. Navigate to working directory
cd /d E:\analysisnifty

:: 2. Prevent committing GCP credentials file
echo service_account.json >> .gitignore

:: 3. Create required workflow folder structure
mkdir .github\workflows

:: 4. Move workflow YAML to correct path
move githubcronjob.yaml .github\workflows\daily_run.yml

:: 5. Re-initialize clean local Git repository
git init
git branch -M main
git add .
git commit -m "Initial commit with proper workflow structure"

:: 6. Connect remote with Personal Access Token (PAT)
git remote add origin https://<PAT_TOKEN>@github.com/TukaramShelar/niftyfifty.git

:: 7. Force push clean repository tree to main branch
```

5. ENCOUNTERED ERRORS, ROOT CAUSES & FIXES

Comprehensive register of every technical issue encountered during development, its underlying cause, and the exact resolution applied:

Error / Symptom	Root Cause	Resolution Action
APIError [403]: Google Drive API has not been used... or disabled	`gsread` uses Google Drive API to search spreadsheet names, but only Sheets API was initially enabled.	Opened GCP Console, enabled Google Drive API for project 547542568504, and re-executed.
Invalid username or token. Password auth not supported	GitHub deprecated password-based Git push operations in August 2021.	Generated a Personal Access Token (PAT) in GitHub Developer Settings and authenticated Git CLI with PAT.
push declined due to repository rule violations	GitHub Push Protection detected `service_account.json` committed in local Git history.	Ran <code>git rm --cached service_account.json</code> , updated <code>.gitignore</code> , cleared <code>.git</code> history, and forced clean commit.
Actions tab showing "Get started with GitHub Actions"	Workflow file was nested incorrectly as <code>.github/workflows/.github/workflows/</code> or placed in root directory.	Created exact directory path <code>.github/workflows/daily_run.yml</code> and cleaned nested folders on main branch.
refusing to allow PAT to update workflow without workflow scope	The generated PAT lacked the explicit `workflow` scope permission required to modify YAML files.	Edited PAT scopes in GitHub settings, checked the `workflow` scope checkbox, and re-executed push.

6. RAPID RE-DEPLOYMENT CHECKLIST FOR FUTURE PROJECTS

1. Create GCP Project, enable **Sheets & Drive APIs**, download Service Account JSON key.
2. Share target Google Sheet with Service Account email address.
3. Create project directory with structure: `.github/workflows/daily_run.yml`, `main.py`, `.gitignore`.
4. Add `service_account.json` to `.gitignore` **BEFORE** running `git add ..`
5. Ensure GitHub Personal Access Token has both **repo** and **workflow** scopes checked.
6. Add GCP_SERVICE_ACCOUNT secret in GitHub Repository Settings `$\rightarrow$ Secrets and variables $\rightarrow$ Actions`.
7. Trigger manual run under **Actions** tab `$\rightarrow$ Run workflow` to verify execution.